Do **not** write solutions on this page.

10. [Maximum mark: 19]

The point $\mathrm{P}(-1, 1, -13)$ lies on the line L_1. The line L_1 has a direction vector $\begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$.

(a) Write down a vector equation for L_1 in the form $\boldsymbol{r} = \boldsymbol{a} + \lambda\boldsymbol{b}$. [1]

(b) Find a vector equation for line L_2 in the form $\boldsymbol{s} = \boldsymbol{c} + \mu\boldsymbol{d}$, given that L_2 passes through the points $\mathrm{A}(2, -4, 2)$ and $\mathrm{B}(7, -6, 1)$. [2]

(c) Show that L_1 and L_2 are skew. [5]

The point N lies on L_2.

(d) Find $\overrightarrow{\mathrm{PN}} \cdot \overrightarrow{\mathrm{AB}}$ in terms of μ. [4]

(e) Given that N is the point on L_2 that lies closest to P, find the coordinates of N. [3]

(f) Point O denotes the origin $(0, 0, 0)$.
Find the equation of the plane containing points O, P and N, giving your answer in the form $\alpha x + \beta y + \gamma z = \delta$, where $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$. [4]